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econ0004 - test

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Econometrics and Data Analysis Test Questions

Part 1: Multiple Choice Questions (35 questions, 2 points each) -16

1. What is econometrics primarily concerned with?

   A. The study of economic history.

   B. The use of statistical methods to analyze economic data.

   C. The development of economic theories without data.

   D. The analysis of financial markets only.

2. Which of the following is an example of time series data?

   A. Data on GDP of a country over 10 years.

   B. Data on household consumption in a single year.

   C. Data on firm productivity across industries in one year.

   D. Data on student test scores across schools.

3. Panel data is characterized by:

   A. Observations on different units over time.

   B. Observations on the same units over time.

   C. A single cross-section of units.

   D. Aggregated national data.

4. Correlation measures:

   A. Causality between two variables.

   B. The degree to which two variables move together.

   C. The percentage change in one variable.

D. The variance of a variable.

5. The statement "correlation does not imply causation" means:

A. Correlation always implies causation.

C

B. Correlation is irrelevant in econometrics.

C. A relationship between variables may be due to a third omitted variable.

D. Causation can be inferred directly from correlation.

6. In the regression model $y_i = \alpha + \beta x_i + \epsilon_i$, ϵ_i represents:

A. The dependent variable.

C

B. The independent variable.

C. The error term.

D. The slope coefficient.

7. The OLS estimator minimizes:

A. The sum of absolute residuals.

B

B. The sum of squared residuals.

C. The variance of the independent variable.

D. The mean of the dependent variable.

8. An estimator is unbiased if:

A. Its expected value equals the true parameter.

A

B. It has the smallest variance.

C. It is consistent.

D. It is efficient.

? 9. Which assumption is required for OLS to be BLUE?

C

A. Heteroskedasticity.

~~A~~

B. Autocorrelation.

C. Homoskedasticity and no autocorrelation.

D. Non-linear parameters.

? 10. Omitted variable bias occurs when:

D

A. The model includes too many variables.

B. A relevant variable is left out of the regression.

C. The error term is correlated with the independent variable.

D. Both B and C.

11. In the wage equation $\ln w_i = \alpha + \beta \text{educ}_i + \epsilon_i$, β represents:

A. The level of wage.

B

B. The percentage change in wage due to one more year of education.

C. The absolute change in wage.

D. The variance of wage.

12. Which of the following is a pooled cross-section?

A. Data on the same firms over time.

B

B. Two independent cross-sections combined.

C. Data on GDP over 20 years.

D. Data on individuals in one year only.

13. Reverse causality refers to:

A. X causing Y.

B

B. Y causing X.

C. Z causing both X and Y.

D. No causal relationship.

14. The ceteris paribus condition is important in causal analysis because:

A. It allows all variables to change.

B

B. It holds other factors constant.

C. It ignores omitted variables.

D. It assumes perfect correlation.

15. In the newspaper readership example, education is considered a:

A. Dependent variable.

~~B~~

B. Independent variable.

C. Confounding variable.

C

D. Error term.

16. The Solow residual measures:

B

A. Capital growth.

B. Productivity growth.

A

C. Labor growth.

D. Output growth.

17. A consistent estimator:

B

A. Is always unbiased.

B. Converges to the true parameter as sample size increases.

C. Has the smallest variance.

D. Is efficient in small samples.

18. Which is NOT a typical goal of econometric analysis?

C

A. Forecasting economic variables.

B. Testing economic theories.

C. Proving economic laws without data.

D. Evaluating policy impacts.

? 19. In the regression $y_i = \alpha + \beta x_i + \epsilon_i$, $\hat{\beta}$ is calculated as:

A

A. Covariance of x and y divided by variance of x.

B. Variance of x divided by covariance of x and y.

C. Mean of y divided by mean of x.

D. Sum of squared residuals.

20. Homoskedasticity means:

A

A. The error term has constant variance.

B. The error term is correlated with x.

C. The error term has zero mean.

D. The independent variable is constant.

? 21. Which data type is best for studying life-cycle phenomena?

B C

A. Cross-sectional data.

B. Time series data.

C. Panel data.

D. Pooled cross-sections.

22. The error term ϵ_i is also called:

B A

A. The regressor.

B. The disturbance term.

C. The slope.

D. The intercept.

23. If $\text{cov}(x_2, x_1) > 0$ and $\beta_2 > 0$, omitting x_2 will cause:

A

A. Positive bias in $\hat{\beta}_1$.

B. Negative bias in $\hat{\beta}_1$.

C. No bias.

D. Bias cannot be determined.

24. In the wage equation, using $\ln w_i$ allows interpretation of β as:

B

A. Absolute change in wage.

B. Percentage change in wage.

C. Level of education.

D. Variance of wage.

25. Which is an example of an omitted variable in the education-wage relationship?

D

A. Years of education.

B. Hourly wage.

C. Ability.

D. Age.

26. The OLS estimator is:

C

A. Always unbiased.

B. Always efficient.

C. BLUE under certain assumptions.

D. Not used in econometrics.

27. A scatter plot of log wage against education shows:

A. A deterministic relationship.

B

B. A stochastic upward trend.

C. No relationship.

D. A perfect line.

28. The normal equations in OLS are derived from:

B

A. Maximizing the likelihood function.

B. Minimizing the sum of squared residuals.

C. Setting residuals to zero.

D. Both B and C.

29. In the newspaper example, if education is controlled for, the relationship between readership and attitudes:

A. Disappears completely.

C

B. Becomes stronger.

C. Persists but may still be confounded.

D. Reverses direction.

30. Which of the following violates OLS Assumption 2 (exogeneity)?

A. Homoskedasticity.

B

B. Omitted variable correlated with x.

C. No autocorrelation.

D. Linear in parameters.

31. The formula for the sample correlation coefficient is: $r_{xy} = \frac{Cov(x,y)}{S_x S_y}$

A. Covariance divided by product of standard deviations.

A

B. Variance of x divided by variance of y.

C. Sum of squared residuals.

D. Mean of x times mean of y.

32. In the production function $y = Ak^{\alpha}h^{1-\alpha}$, A represents:

C

A. Capital per worker.

B. Human capital.

C. Total factor productivity.

D. Output per worker.

33. Pooled cross-sections are often used to:

A. Study time trends.

B

B. Evaluate policy changes.

C. Analyze panel data.

D. Forecast GDP.

34. The intercept α in the regression line represents:

A. The slope of the line.

B

B. The value of y when $x = 0$.

C. The error term.

D. The covariance.

35. If $E(\epsilon_i|x_i) \neq 0$, which OLS assumption is violated?

A. Linearity.

BD

B. Exogeneity.

C. Homoskedasticity.

D. No autocorrelation.

36. In the growth accounting equation, the Solow residual is calculated as:

A. $\hat{A} = \hat{y} - \alpha\hat{k} - (1-\alpha)\hat{h}$.

AB

B. $\hat{y} = \hat{A} + \alpha\hat{k} + (1-\alpha)\hat{h}$.

C. $\hat{k} = \hat{y}/\hat{A}$.

D. $\hat{h} = \hat{y} - \hat{A}$.

Part 2: Multiple Select Questions (5 questions, 2 points each) -1

1. Which of the following are typical goals of econometric analysis? (Select all that apply.)

A. Estimating relationships between economic variables.

ACD

B. Proving economic theories without data. X

C. Forecasting economic variables.

D. Evaluating government policies.

2. Which of the following statements about correlation and causation are true? (Select all that apply.)

A. Correlation always implies causation. X

BCD

B. Causation implies correlation.

C. Omitted variables can create spurious correlation.

D. Reverse causality is a form of endogeneity.

3. Which of the following are types of data used in applied economics? (Select all that apply.)

A. Time series data.

ABC

B. Cross-sectional data.

C. Panel data.

D. Experimental data only.

4. Which of the following are desirable properties of an estimator? (Select all that apply.)

A. Unbiasedness.

ABC

B. Consistency.

C. Efficiency.

D. Complexity.

5. Which of the following can cause OLS Assumption 2 (exogeneity) to fail? (Select all that apply.)

A. Omitted variable correlated with x.

AB C

B. Reverse causality.

C. Measurement error in x.

D. Homoskedasticity.

Part 3: True/False Questions (10 questions, 2 points each) −2

1. Econometrics is the use of statistical methods to analyze economic data. (True/False)

2. Panel data consists of observations on different units over time. (True/False)

3. Correlation implies causation. (True/False)

4. The OLS estimator minimizes the sum of absolute residuals. (True/False)
5. An unbiased estimator always gives the correct estimate in a single sample. (True/False)
6. Homoskedasticity means the error term has constant variance. (True/False)
7. The ceteris paribus assumption is irrelevant in causal analysis. (True/False)
8. Omitted variable bias occurs when an irrelevant variable is included in the model. (True/False)
9. The error term in a regression model represents unobserved factors. (True/False)
10. In the wage equation, β represents the absolute change in wage for an additional year of education. (True/False)